DATA ANALYTICS CAPSTONE PROJECT

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NETFLIX

About the Netflix

Netflix is a leading global streaming and entertainment service that provides a vast library of movies, TV series, documentaries, and award-winning original content across multiple languages and genres. Founded in 1997 in the United States, Netflix initially operated as a DVD-by-mail rental service before transitioning into online streaming in 2007. This shift revolutionized how people consume entertainment.

Netflix operates on a subscription-based model, allowing users to watch content anytime and anywhere on devices such as smartphones, smart TVs, tablets, laptops, and gaming consoles. One of its key strengths is its strong focus on original programming, including popular shows and films like *Stranger Things*, *Money Heist*, *The Crown*, and *Squid Game*.

Today, Netflix is available in more than 190 countries and continues to play a major role in shaping the future of digital entertainment. It has changed traditional television viewing habits, encouraged binge-watching, and influenced how content is produced, marketed, and consumed worldwide.

Aim

The primary aim of this project is to perform Exploratory Data Analysis (EDA) on the Netflix dataset to gain meaningful insights into the platform's content library. This project seeks to understand the structure, distribution, trends, and patterns of movies and TV shows available on Netflix by analysing various attributes such as content type, genres, release years, duration, ratings, countries, and languages. Through systematic data cleaning, transformation, and visualization, the project aims to identify how Netflix's content has evolved over time, how it is distributed across different regions, and how different genres and categories contribute to content diversity. The analysis also focuses on uncovering relationships between variables—such as ratings and duration—to better understand content characteristics. Additionally, this project aims to explore user preference indicators, such as the popularity of specific genres or content types, and to evaluate Netflix's strategy in terms of content variety, geographical reach, and audience targeting. By leveraging visual analytics using tools like Plotly, the project transforms raw data into clear, interpretable insights that support data-driven conclusions and recommendations.

Ultimately, the goal of this EDA project is to enhance analytical thinking, strengthen data visualization skills, and demonstrate how exploratory analysis can be used to extract actionable insights from real-world datasets.

Objective

The objective of this project is to analyse the Netflix Titles dataset to gain insights into the platform's content library, including movies and TV shows. This project aims to understand content distribution based on genre, country, release year, rating, and duration. By exploring trends in Netflix's content strategy, the analysis helps identify patterns such as growth in original content, popular genres, and regional contributions.

Additionally, the project focuses on using data analysis and visualization techniques to transform raw data into meaningful insights that support data-driven decision-making. The findings can assist stakeholders in understanding viewer preferences, content trends, and Netflix's global expansion strategy.

Tools

Python

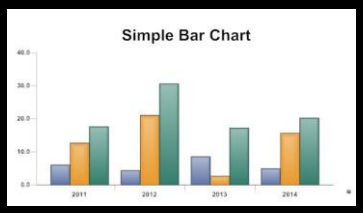
Use:

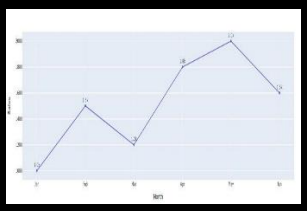
A high-level programming language widely used for data analysis, automation, and AI. Features: Easy to learn and read. Huge library support for data, web, and AI.

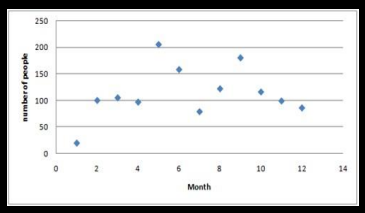
Plotly

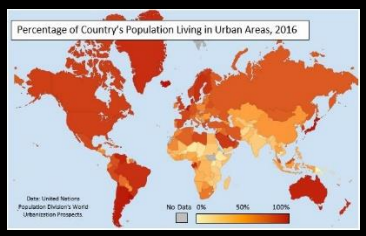
Plotly is an interactive data visualization library used to create dynamic and engaging charts. It supports various visualizations such as bar charts, line charts, scatter plots, pie charts, and maps. Plotly allows users to interact with data through features like zooming, hovering, and filtering.

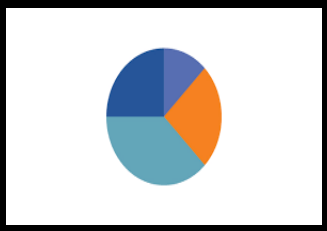
Graphs

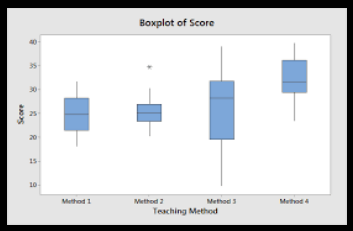












Problem Statments

TASKS OVERVIEW -

TASK 1 - BEGIN BY LOADING AND EXPLORING THE DATASET.

TASK 2 - CLEAN AND PREPROCESS DATA TO ENSURE CONSISTENCY AND ACCURACY FOR ANALYSIS.

TASK 3 - COMPUTE BASIC DESCRIPTIVE STATISTICS SUCH AS MEAN, MEDIAN, MODE, RANGE, AND STANDARD DEVIATION FOR RELEVANT.

TASK 4 - CREATE VISUALIZATIONS TO REPRESENT THE DISTRIBUTION OF CONTENT OVER DIFFERENT GENRES.

TASK 5 - VISUALIZE THE DISTRIBUTION OF CONTENT ACROSS RELEASE YEARS.

TASK 6 - EXPLORE THE GEOGRAPHICAL DISTRIBUTION OF CONTENT (IF APPLICABLE).

TASK 7 - IF THERE'S A TEMPORAL COMPONENT, PERFORM TIME SERIES ANALYSIS TO IDENTIFY TRENDS AND PATTERNS OVERTIME.

TASK 8 - ANALYZE THE DISTRIBUTION OF CONTENT RATINGS.
TASK 9 - EXPLORE THE LENGTH OF MOVIES OR EPISODES AND IDENTIFY ANY TRENDS.

TASK 10 - IDENTIFY AND PRESENT TOP-RATED MOVIES OR TV SHOWS BASED ON USER RATINGS.

TASK 11 - ANALYZE TRENDS IN THE POPULARITY OF DIFFERENT GENRES OVER TIME.

TASK 12 - FURTHER EXPLORE THE DISTRIBUTION OF CONTENT ACROSS DIFFERENT COUNTRIES AND REGIONS.

TASK 13 - INVESTIGATE POTENTIAL CORRELATIONS BETWEEN VARIABLES (E.G., RATINGS AND DURATION).

TASK 14 - EVALUATE THE DIVERSITY OF CONTENT BY ANALYZING THE NUMBER OF UNIQUE GENRES AND CATEGORIES.

TASK 15 - EXPLORE HOW THE CHARACTERISTICS OF CONTENT (E.G., DURATION, RATINGS) HAVE EVOLVED OVER THE YEARS.

Data Overview

In this project, **Pandas** is used for data handling and analysis, **NumPy** for numerical operations, **Matplotlib** and **Seaborn** for creating visualizations, and **Plotly** for interactive charts.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly as px
```

Pd.read_csv is used to import the dataset from a CSV file into a Pandas DataFrame, allowing easy data viewing, cleaning, and analysis.

```
a = pd.read_csv("/content/netflix_titles.csv")
```

```
a
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...
...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...

Data Cleaning

```
a.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

The info() function provides a summary of the Netflix dataset, showing 8,807 records and 12 columns along with their data types and non-null values. It helps identify missing data, understand the structure of the dataset, and estimate memory usage for efficient analysis.

```
a.shape
```

```
(8807, 12)
```

We use the shape() function to quickly understand the dataset's size. The dataset contains 8,807 records and 12 attributes.

```
a.describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

We applied the describe() function to summarize key statistics for numerical columns.

```
a.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

The table shows the count of missing values in each column of the dataset. Columns like director, cast, and country have a significant number of missing values (2634, 825, and 831 respectively), while columns such as show_id, type, title, release_year, listed_in, and description have no missing values.

Handling Missing Values

```
a['director'].fillna("Unknown",inplace = True)
```

```
/tmp/ipython-input-637188048.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the

  a['director'].fillna("Unknown",inplace = True)
```

```
a.cast.fillna("Unknown", inplace = True)
```

```
/tmp/ipython-input-2021634509.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the

  a.cast.fillna("Unknown", inplace = True)
```

```
a.country.fillna("Unknown", inplace = True)
```

```
/tmp/ipython-input-2630751910.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the

  a.country.fillna("Unknown", inplace = True)
```

```
a.rating.fillna("Unknown", inplace = True)
```

```
/tmp/ipython-input-3321789927.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the
```

```
a.rating.fillna("Unknown", inplace = True)
```

```
a.duration.fillna("a.rating", inplace = True)
```

```
/tmp/ipython-input-1866264820.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the
```

```
a.duration.fillna("a.rating", inplace = True)
```

```
a[a["date_added"].isnull()]
```

```
a.loc[6066, "date_added"] = "october 2,2013"
a.loc[6174, "date_added"] = "April 14, 2013"
a.loc[6795, "date_added"] = "january 15, 1993"
a.loc[6806, "date_added"] = "September 22, 1994"
a.loc[6901, "date_added"] = "july 23, 2003"
a.loc[7196, "date_added"] = "august 14, 2004"
a.loc[7254, "date_added"] = "october 5, 2002"
a.loc[7406, "date_added"] = "May 3, 2013"
a.loc[7847, "date_added"] =  "june 18, 2003"
a.loc[8182, "date_added"] = "may 13, 2012"
```

```
a.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0
description	0

dtype: int64

All missing values in the dataset were successfully handled through appropriate data cleaning techniques, ensuring data completeness and consistency for accurate analysis and modelling.

Checking Duplicates

```
a.duplicated().sum()
```

```
np.int64(0)
```

There is no duplicate in the dataset.

Insights

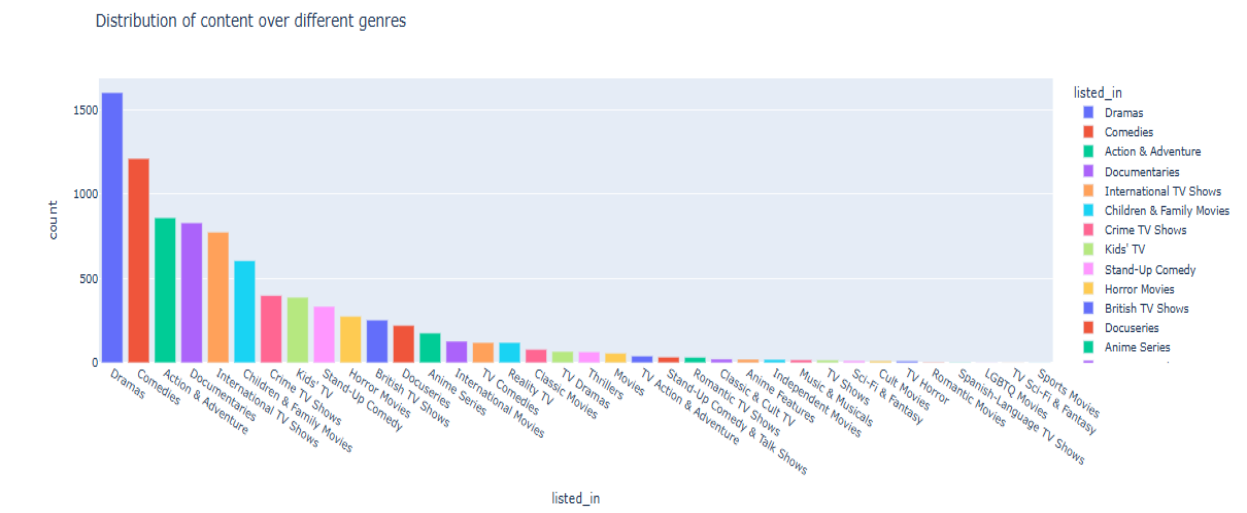
DISTRIBUTION OF CONTENT OVER DIFFERENT GENRES

```
a.listed_in = a.listed_in.str.split(',').str[0]
```

```
dd = a.listed_in.value_counts().reset_index()
dd
```

	listed_in	count
0	Dramas	1600
1	Comedies	1210
2	Action & Adventure	859
3	Documentaries	829
4	International TV Shows	774
5	Children & Family Movies	605
6	Crime TV Shows	399
7	Kids' TV	388
8	Stand-Up Comedy	334
9	Horror Movies	275
10	British TV Shows	253
11	Docuseries	221

```
px.bar(dd, x = "listed_in", y = "count", color = "listed_in", title = "Distribution of content over different genres")
```



Overview:

The chart represents the distribution of content across different genres, showing the number of titles available in each category on the platform.

Outcome:

Dramas lead with the highest content count, followed by Comedies and Action & Adventure, indicating a strong focus on popular genres, while niche categories have comparatively fewer titles.

DISTRIBUTION OF CONTENT ACROSS RELEASE YEAR

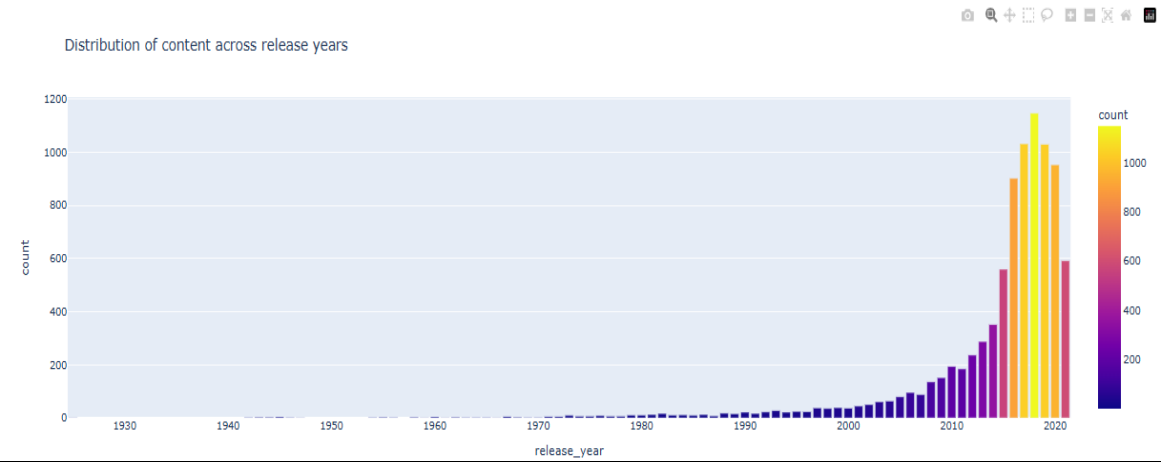
```
df = a.release_year.value_counts().reset_index()
```

```
df
```

	release_year	count
0	2018	1147
1	2017	1032
2	2019	1030
3	2020	953
4	2016	902
...	...	...
69	1961	1
70	1925	1
71	1959	1
72	1966	1
73	1947	1

74 rows × 2 columns

```
px.bar(df, x = 'release_year', y = 'count' , color = "count", title = " Distribution of content across release years")
```



Overview:
The graph shows the distribution of content based on release years, highlighting how the number of titles has changed over time.

Outcome:
Content releases increase gradually in earlier years and rise sharply after 2010, with peak releases in recent years. This indicates rapid growth and a strong focus on expanding content in the modern streaming era.

GEOGRAPHICAL DISTRIBUTION OF CONTENT

```
b = a.country.value_counts().reset_index()
b
```

	country	count
0	United States	2818
1	India	972
2	United Kingdom	419
3	Japan	245
4	South Korea	199
...	...	...
743	Mexico, United States, Spain, Colombia	1
744	Canada, Norway	1
745	Finland, Germany, Belgium	1
746	Argentina, United States, Mexico	1
747	United Kingdom, United States, Germany, Denmar...	1

748 rows × 2 columns

```
c= px.choropleth(b,locations= b["country"],locationmode="country names",
                 projection = "orthographic",color = b['count'],title=
                 " geographical distribution of content")
c
```

Overview:

The choropleth map displays the geographical distribution of content across different countries. It visually represents the volume of content produced or available in each country using color intensity. Darker shades indicate higher content counts, while lighter shades represent lower contributions. This helps identify regional content concentration worldwide.

Outcome:

The map shows that content is heavily concentrated in a few countries, while many regions have comparatively lower representation. This indicates an uneven global distribution of content. Overall, the platform has a strong presence in major content-producing countries, with scope for expansion in underrepresented regions.

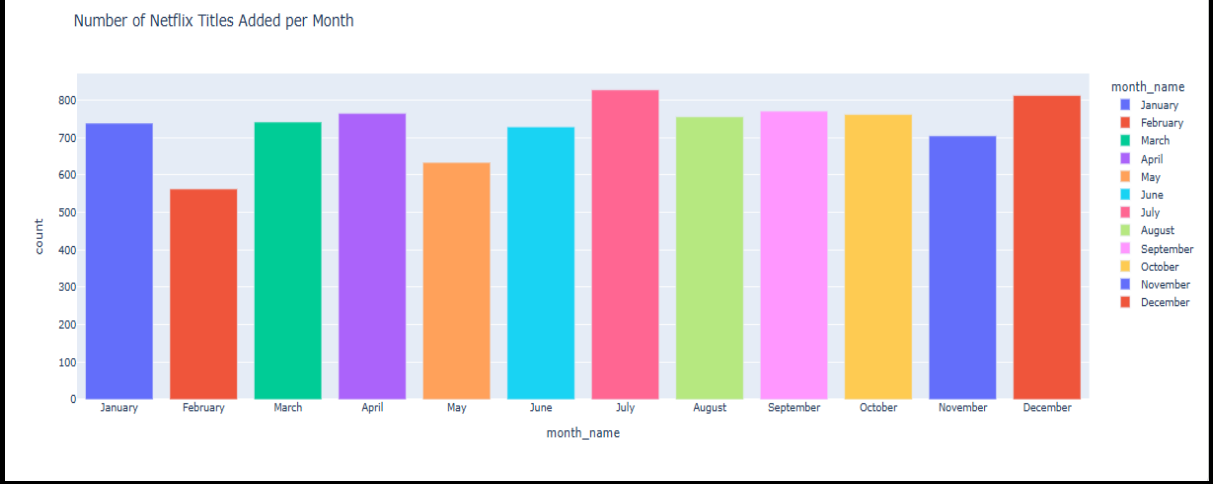
NUMBER OF NETFLIX TITLE ADDED PER MONTH

```
a['date_added'] = pd.to_datetime(a['date_added'], format='mixed')
a['month_name'] = a['date_added'].dt.month_name()
```

```
a['month_name']
month_order = ['January','February','March','April','May','June','July','August','September','October','November','December']
```

```
a['month_name'] = pd.Categorical(
    a['month_name'],
    categories=month_order,
    ordered=True)
d = a.month_name.value_counts().sort_index().reset_index()
```

```
px.bar(d, x = 'month_name', y = 'count', color = "month_name", title = "Number of Netflix Titles Added per Month")
```



Overview:

The bar chart illustrates the number of Netflix titles added in each month throughout the year. It compares monthly content additions to identify seasonal trends in content release. Each bar represents the total titles added in a specific month. This visualization helps understand monthly release patterns.

Outcome:

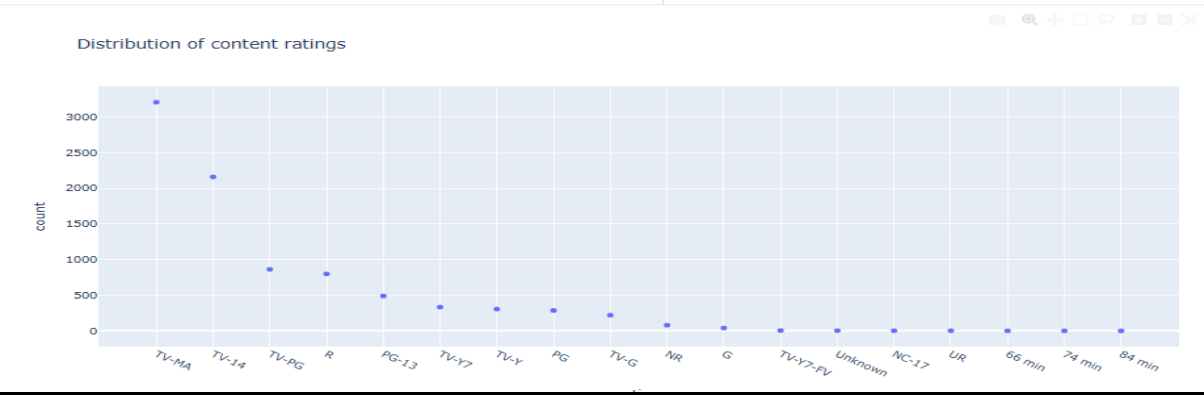
Content additions remain fairly consistent across most months, with noticeable peaks in **July and December**. February and May show comparatively lower additions. Overall, Netflix appears to release more content during mid-year and year-end periods, likely to align with higher viewer engagement.

DISTRIBUTION OF CONTENT RATING

```
r = a.rating.value_counts().reset_index()
r
```

	rating	count
0	TV-MA	3207
1	TV-14	2160
2	TV-PG	863
3	R	799
4	PG-13	490
5	TV-Y7	334
6	TV-Y	307
7	PG	287
8	TV-G	220
9	NR	80
10	G	41
11	TV-Y7-FV	6
12	Unknown	4
13	NC-17	3
14	UR	3
15	66 min	1
16	74 min	1

```
px.scatter(r , x = "rating", y = "count", title = "Distribution of content ratings")
```



Overview

This scatter plot shows the distribution of content ratings by count. It compares different rating categories (like TV-MA, TV-14, PG-13, etc.) against how frequently they appear, giving a quick snapshot of which ratings dominate the dataset.

Outcome

The data clearly indicates that **TV-MA and TV-14** content is most prevalent, suggesting a strong focus on mature and teen audiences. In contrast, kids-friendly and niche ratings (such as TV-Y7-FV, NC-17, or very short-duration categories) have significantly lower counts, highlighting limited content availability in those segments.

AVERAGE DISTRIBUTION OF MOVIE DURATION PER YEAR

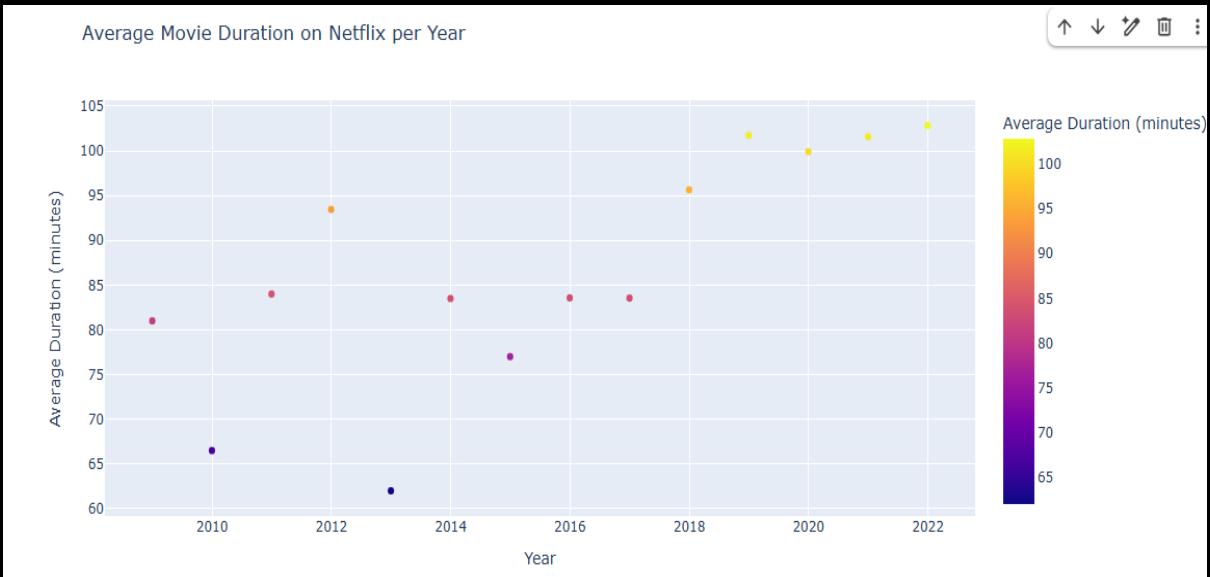
```
movies = df[df["type"] == "Movie"].copy()
movies["duration_min"] = movies["duration"].str.replace(" min", "").astype(int)
movies["date_added"] = pd.to_datetime(movies["date_added"])
avg_duration = ( movies.set_index("date_added").resample("YE")["duration_min"].mean())
```

```
f = (movies.set_index("date_added").resample("YE")["duration_min"].mean().reset_index())
```

```
f
```

	date_added	duration_min
0	2008-12-31	81.000000
1	2009-12-31	66.500000
2	2010-12-31	84.000000
3	2011-12-31	93.461538
4	2012-12-31	62.000000
5	2013-12-31	83.500000
6	2014-12-31	77.000000
7	2015-12-31	83.571429
8	2016-12-31	83.545455
9	2017-12-31	95.654350
10	2018-12-31	101.739693
11	2019-12-31	99.923455

```
px.scatter(f, x="date_added", y="duration_min",
    title="Average Movie Duration on Netflix per Year",
    labels={"date_added": "Year",
        "duration_min": "Average Duration (minutes)" },color = "duration_min")
```



Overview

This graph illustrates the trend of **average movie duration on Netflix over the years**, showing how the typical length of movies has changed from earlier years to recent ones.

Outcome

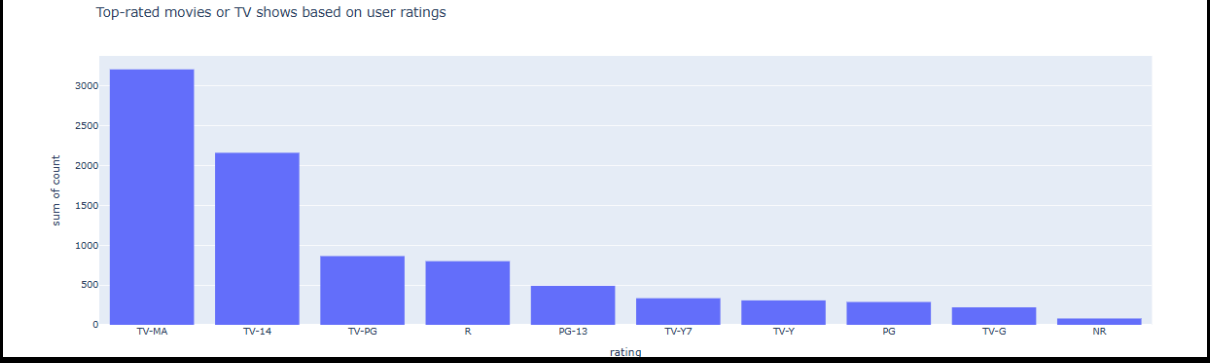
The data reveals a **clear upward trend in average movie duration**, especially after 2018. This suggests that Netflix has increasingly focused on **longer-form movies**, reflecting a shift toward more detailed storytelling and cinematic content in recent years.

TOP RATED MOVIES AND TV SHOWS ON NETFLIX

```
g = a.rating.value_counts().reset_index()
g
```

	rating	count
0	TV-MA	3207
1	TV-14	2160
2	TV-PG	863
3	R	799
4	PG-13	490
5	TV-Y7	334
6	TV-Y	307
7	PG	287
8	TV-G	220
9	NR	80
10	G	41
11	TV-Y7-FV	6
12	Unknown	4
13	NC-17	3
14	UR	3
15	66 min	1

```
px.histogram(g.head(10), x = "rating", y = "count", title = " Top-rated movies or TV shows based on user ratings")
```



Overview

This bar chart displays the **top-rated movies and TV shows based on user ratings**, comparing different content rating categories with their total counts.

Outcome

The results show that **TV-MA and TV-14** dominate user-rated content, indicating higher availability and popularity of mature and teen-focused titles. Family-friendly and unrestricted categories (such as TV-G and NR) have much lower counts, suggesting that user engagement is strongest with **more mature-rated content** on the platform.

TREND IN POPULARITY OF NETFLIX GENRES OVER TIME

```
a['date_added'] = pd.to_datetime(a["date_added"], format = "mixed")
a['year_added'] = a['date_added'].dt.year
a.listed_in = a.listed_in.str.split(',').str[0]
```

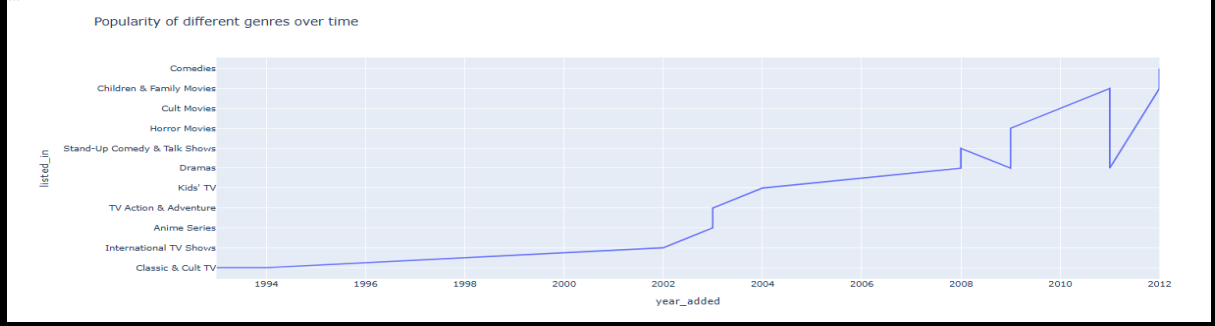
```
h = (a.groupby(['year_added', 'listed_in']).size().reset_index(name='count'))
```

```
h
```

	year_added	listed_in	count
0	1993	Classic & Cult TV	1
1	1994	Classic & Cult TV	1
2	2002	International TV Shows	1
3	2003	Anime Series	1
4	2003	TV Action & Adventure	1
...	...	...	...
232	2021	TV Comedies	27
233	2021	TV Dramas	13
234	2021	TV Horror	3
235	2021	TV Shows	5
236	2021	Thrillers	22

237 rows × 3 columns

```
px.line(h.head(15), x = "year_added", y = "listed_in", title = "Popularity of different genres over time" )
```



Overview

This line chart shows the **popularity of different content genres over time**, based on the year titles were added and their listed genres on the platform.

Outcome

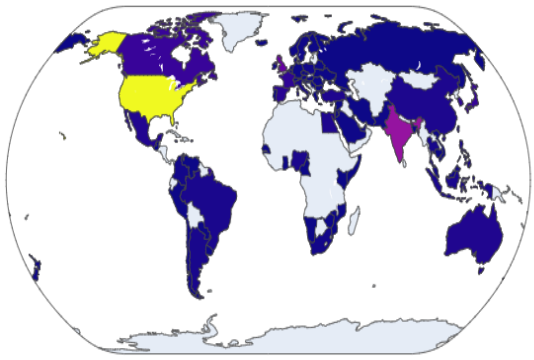
The trend indicates a **gradual expansion and diversification of genres** over the years. In later periods, there is a noticeable rise in categories such as **TV Action & Adventure, International TV Shows, Anime Series, and Kids' TV**, suggesting that the platform increasingly focused on **global, varied, and niche audience preferences** rather than a limited set of genres.

DISTRIBUTION OF CONTENT ACROSS DIFFERENT COUNTRIES AND REGIONS

```
a.country = a.country.str.split(',').str[0]
```

```
i = a.country.value_counts().reset_index()
```

```
px.choropleth(i,locations= i["country"],locationmode="country names",projection = "natural earth",color = i['count'])
```



Overview

This choropleth world map visualizes the **country-wise distribution of content**, where colour intensity represents the total count of titles available in each country.

Outcome

The map highlights a **strong concentration of content in a few key countries**, with the **United States leading by a wide margin**, followed by countries like **India and parts of Europe**. Many regions show comparatively lower counts, indicating that content production and availability are **unevenly distributed globally**, with a primary focus on major media markets.

POTENTIAL CORRELATIONS BETWEEN VRIABLES (E.G.,RATINGS AND DURATIONS)

```
movies = df[df["type"] == "Movie"].copy()
movies["duration_min"] = movies["duration"].str.replace(" min", "").astype(int)
```

```
df["rating"].unique()
```

```
array(['PG-13', 'TV-MA', 'PG', 'TV-14', 'TV-PG', 'TV-Y', 'TV-Y7', 'R',
       'TV-G', 'G', 'NC-17', '74 min', '84 min', '66 min', 'NR',
       'TV-Y7-FV', 'UR'], dtype=object)
```

```
rating_order = {
    'G': 1,
    'PG': 2,
    'PG-13': 3,
    'R': 4,
    'NC-17': 5,
    'TV-Y': 1,
    'TV-Y7': 2,
    'TV-G': 2,
    'TV-PG': 3,
    'TV-14': 4,
    'TV-MA': 5}

j = movies['rating_numeric'] = movies['rating'].map(rating_order)
```

```
px.box(movies,x='rating_numeric',y='duration_min',
    title='Distribution of Movie Duration Across Ratings',color = "rating_numeric")
```

Overview

This box plot illustrates the **distribution of movie durations across different rating categories**, showing how movie length varies by rating level, including median, spread, and outliers.

Outcome

The results indicate that **higher-rated categories generally have longer and more consistent movie durations**, with ratings 3 and 4 showing higher median runtimes. Lower-rated categories display wider variability and shorter typical durations, suggesting that **better-rated content tends to be more standard in length and often longer in runtime**.

ANALYZING THE NUMBER OF UNIQUE GENRES AND CATEGORIES

UNIQUE GENRES-

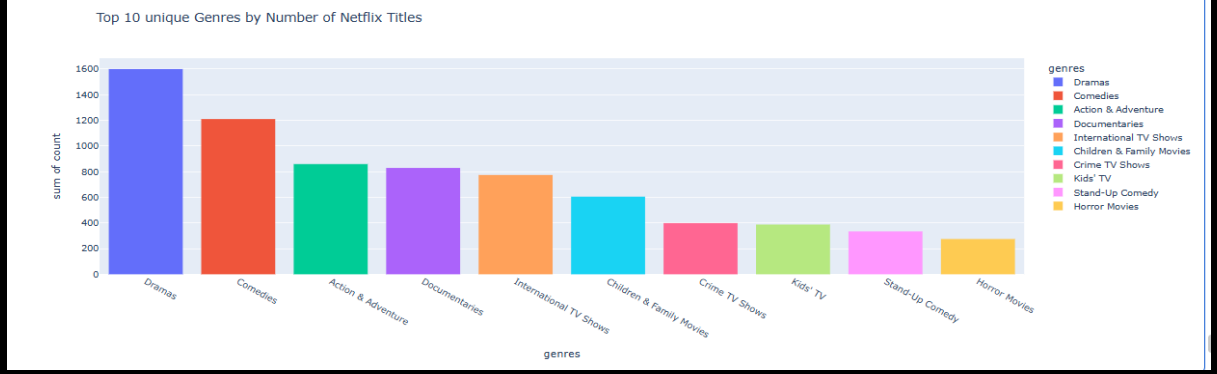
```
df.listed_in = df.listed_in.str.split(',').str[0]

k = df.listed_in.value_counts().reset_index()

k
```

4	International TV Shows	774
5	Children & Family Movies	605
6	Crime TV Shows	399
7	Kids' TV	388
8	Stand-Up Comedy	334
9	Horror Movies	275
10	British TV Shows	253
11	Docuseries	221
12	Anime Series	176
13	International Movies	128
14	TV Comedies	120

```
px.histogram(k.head(10),x='listed_in',y='count',color = "listed_in",labels = {"listed_in": " genres"},title='Top 10 unique Genres by Number of Netflix Titles')
```



Overview

This bar chart highlights the **top 10 unique genres on Netflix**, ranked by the total number of titles available in each genre.

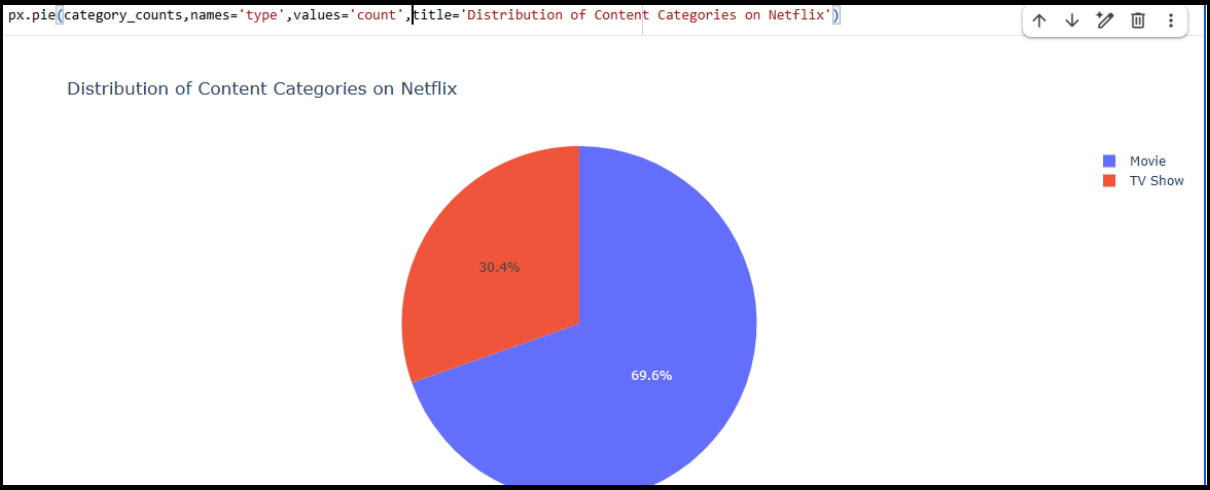
Outcome

The visualization shows that **Dramas and Comedies dominate Netflix's content library**, followed by **Action & Adventure and Documentaries**. Niche genres such as **Stand-Up Comedy and Horror Movies** have comparatively fewer titles, indicating that Netflix's catalog is **heavily focused on mainstream, high-demand genres** while still maintaining some diversity.

CONTENT CATEGORIES-

```
category_counts = (df['type'].value_counts().reset_index())
category_counts.columns = ['type', 'count']
```

```
px.pie(category_counts,names='type',values='count',title='Distribution of Content Categories on Netflix')
```



Overview

This pie chart represents the **distribution of content categories on Netflix**, comparing the share of **Movies** and **TV Shows** in the overall catalog.

Outcome

The chart clearly shows that **Movies make up the majority of Netflix's content (around 70%)**, while **TV Shows account for roughly 30%**. This indicates that Netflix's library is **movie-heavy**, though TV shows still form a significant and strategic portion of the platform's offerings.

EXPLORING THE CHARACTERISTICS OF CONTENT (E.g., DURATION, RATINGS) OVER THE YEARS

Average duration of Netflix movies over time

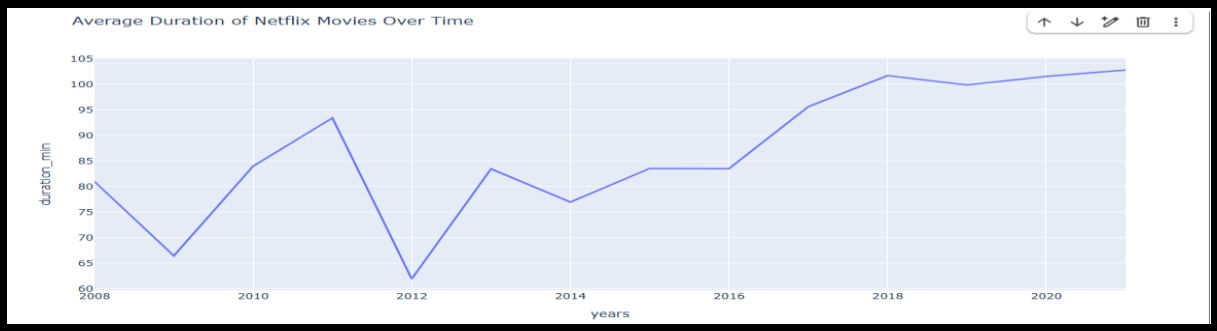
```
movies = df[df["type"] == "Movie"].copy()
movies["duration_min"] = movies["duration"].str.replace(" min", "").astype(int)
movies["date_added"] = pd.to_datetime(movies["date_added"])
df['year_added'] = df['date_added'].dt.year
```

```
M= (movies.groupby('year_added')['duration_min'].mean().reset_index())
```

```
M
```

	year_added	duration_min
0	2008	81.000000
1	2009	66.500000
2	2010	84.000000
3	2011	93.461538
4	2012	62.000000
5	2013	83.500000
6	2014	77.000000

```
px.line(M,x='year_added',y='duration_min',labels = {"year_added" : "years"},title='Average Duration of Netflix Movies Over Time')
```



Overview

This line chart illustrates the **trend in average duration of Netflix movies over time**, showing how movie runtimes have changed across different years.

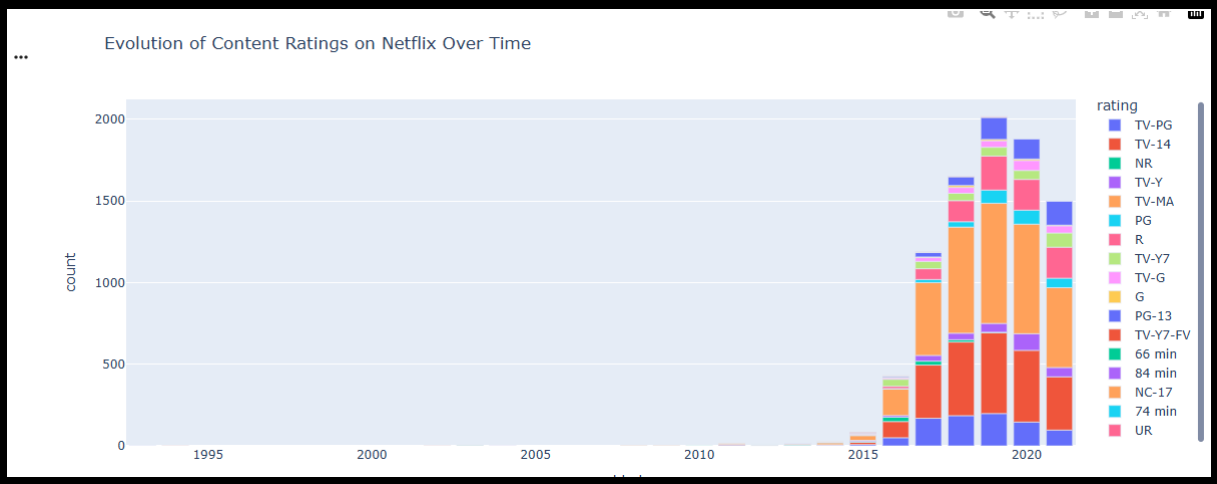
Outcome

The graph reveals an **overall increase in average movie duration**, especially after 2016. Despite some early fluctuations, recent years show longer runtimes, indicating Netflix's shift toward **more in-depth, feature-length storytelling** and higher production value content.

EVOLUYION OF CONTENT RATINGS ON NETFLIX OVER TIME

```
rating_trend = (df.groupby(['year_added', 'rating']).size().reset_index(name='count'))

# # TASK 15 :- EXPLORE HOW THE CHARACTERISTICS OF CONTENT (E.G., DURATION, RATINGS) HAVE EVOLVED OVER THE YEARS.
import plotly.express as pxpx.bar(rating_trend,x='year_added',y='count',color='rating',title='Evolution of Content Ratings on Netflix Over Time')
```



Overview

This stacked bar chart shows the **evolution of content ratings on Netflix over time**, highlighting how different rating categories have contributed to the total content count each year.

Outcome

The graph indicates a **sharp growth in content after 2015**, with **TV-MA and TV-14** becoming increasingly dominant. This suggests that Netflix has progressively shifted its focus toward **mature and teen-oriented content**, while kids and general-audience ratings have grown at a slower pace.

Conclusion

This exploratory data analysis of the Netflix dataset provides a comprehensive understanding of Netflix's content strategy, growth patterns, and audience targeting over time. The dataset, consisting of over 8,800 titles, was carefully cleaned by handling missing values in key fields such as director, cast, country, date added, rating, and duration, ensuring reliable and consistent analysis.

A) Content Composition & Categories

The analysis reveals that Movies dominate Netflix's catalog (≈70%), while TV Shows account for around 30%, highlighting Netflix's stronger focus on standalone content while still maintaining a substantial episodic library. This balance allows Netflix to cater to both binge-watchers and casual viewers.

B) Genre Distribution Insights

Genre-wise exploration shows that Dramas and Comedies are the most prevalent genres, followed by Action & Adventure, Documentaries, and International TV Shows. This indicates Netflix's emphasis on emotionally driven storytelling and light entertainment, while also investing heavily in global and region-specific content, reflecting its international expansion strategy. Niche genres such as LGBTQ Movies, Sports Movies, and Sci-Fi TV content exist but form a very small fraction of the catalogue.

C) Temporal & Growth Trends

The release year and date-added analysis highlight a sharp rise in content additions after 2015, peaking between 2017 and 2020, which aligns with Netflix's aggressive global growth and original content push. Monthly trends show that Netflix consistently adds content throughout the year, with noticeable spikes in certain

months, suggesting strategic release planning to maximize viewer engagement.

D) Geographical Distribution

Geographical analysis confirms that the United States is the largest content producer, followed by India and the United Kingdom. The strong presence of Indian content reflects Netflix's growing focus on emerging markets and regional storytelling. Overall, the platform demonstrates wide global coverage, reinforcing its identity as a truly international streaming service.

E) Ratings & Audience Targeting

Content rating analysis indicates a clear skew toward mature audiences, with TV-MA and TV-14 being the most common ratings. This suggests Netflix prioritizes content aimed at young adults and adults, while still maintaining a reasonable catalog for children and family audiences through TV-Y and TV-G rated content.

F) Duration & Content Evolution

The analysis of movie duration over time reveals a gradual increase in average movie length, especially after 2016. This trend suggests a shift toward more in-depth storytelling and higher production value films. Additionally, the relationship between ratings and duration shows that mature-rated movies generally have longer runtimes, indicating more complex narratives.

Acknowledgement

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Finally, I appreciate the constructive feedback and suggestions from my classmates, which played a crucial role in refining my project at various stages.

The successful completion of this project was a collective effort, and I am deeply appreciative of the time and care invested by everyone. This learning experience has been incredibly fruitful, and I will always remember the collaborative efforts that made it possible.

THANKYOU